



江苏华起微电子有限公司
GENOVA MICROTECH CO., LTD

HQEF5016 Datasheet

16V,50A Electronic Fuse

Rev.1.1

HQEF5016

16V,50A Electronic Fuse

DESCRIPTION

The HQEF5016 is a single-channel current-limited switch that integrates a MOSFET and a controller. It has a low $R_{DS(ON)}$ of $1m\Omega$, and a single HQEF5016 can handle continuous currents up to 50A. It also supports parallel operation for applications requiring higher currents.

The HQEF5016 can limit the high inrush current generated during hot-plug scenarios, preventing a significant drop in the power bus voltage. It provides high-precision overcurrent protection, ensuring effective circuit protection.

The HQEF5016 supports programmable soft-start time. It provides an integrated current mirror function for accurate current reporting. It includes temperature monitoring capability, eliminating the need for an external temperature detection circuit. The HQEF5016 also supports fault indication, allowing clear visibility into the device's operational status.

The HQEF5016 also features robust protection capabilities, including under-voltage protection, over-voltage protection, over-current protection, short-circuit current limit, MOSFET fault diagnosis protection, over-temperature protection, and 3.3V LDO current limit protection.

The HQEF5016 is available in an LGA-32 (5mm*5mm) package.

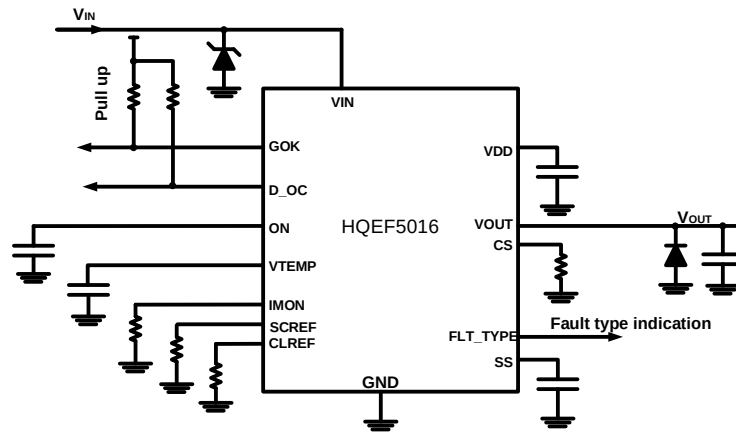
FEATURES

- 4V to 16V Operating Input Range
- Maximum 50A Output Current
- $1m\Omega$ On-Resistance
- Supporting Transient Over-Current as a Circuit Breaker
- 3.3V LDO Output with Current Limit
- Integrated Current Sensing with Sense Output
- $10\mu A/A$ Current Sensing Analog Output
- Built-In Insertion Delay
- Adjustable Soft-Start
- Output Short-Circuit Protection
- Over-Voltage Protection
- Over-Temperature Protection
- Fault Signal Output
- Parallel Operation for High-Current Application
- $8.7mV/^{\circ}C$ Temperature Analog Output
- Available in LGA-32 (5mm*5mm) Package

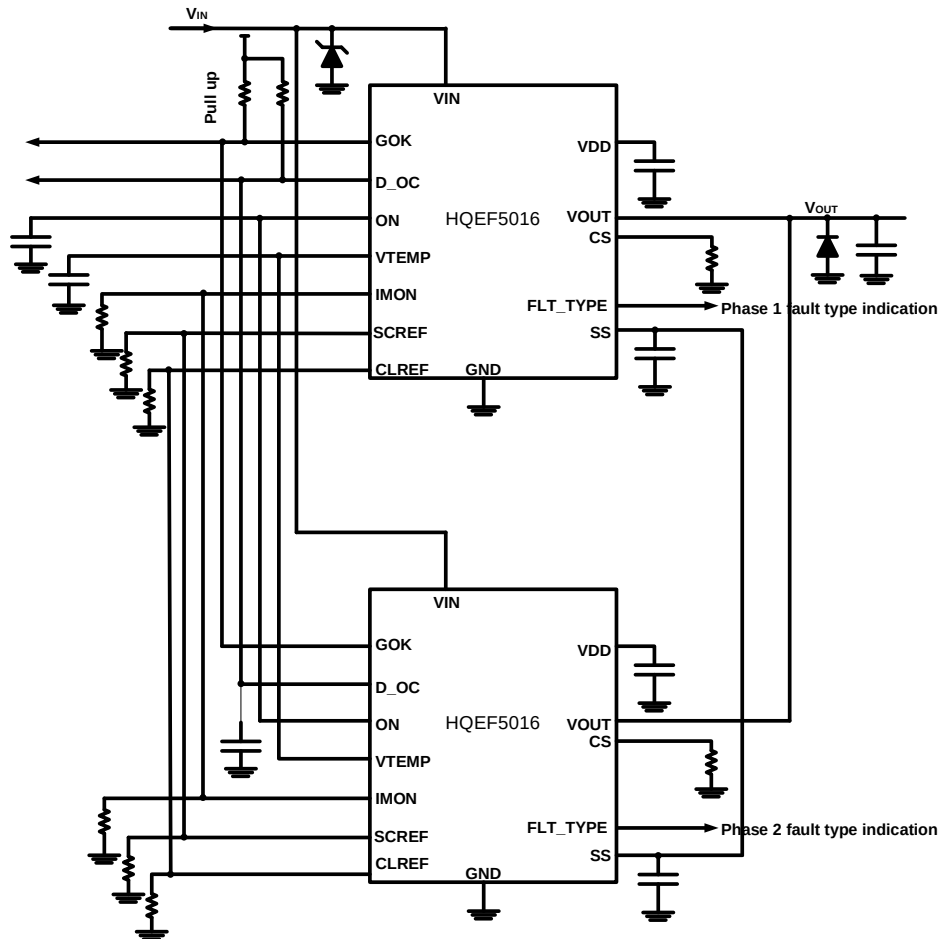
APPLICATIONS

- Server
- Networking
- Plug-In Modules

TYPICAL APPLICATION



Standalone Operation



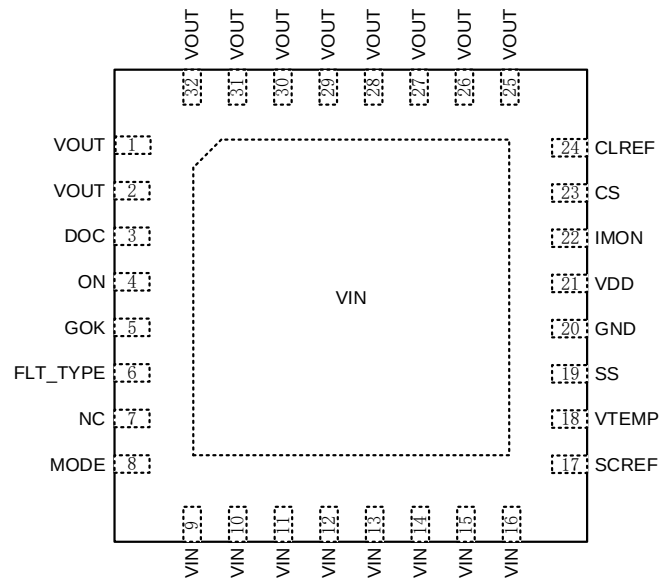
Parallel Operation

ORDER INFORMATION

DEVICE ¹⁾	QUALITY GRADE	PACKAGE	TOP MARKING ²⁾	ENVIRONMENTAL ³⁾
HQUEF5016	工业级	LGA5*5-32	HQUEF5016 YW□□□□□	Green
HQUEF5016M	普军级	LGA5*5-32	HQUEF5016M YW□□□□□	Green
HQUEF5016J	军温级	LGA5*5-32	HQUEF5016J YW□□□□□	Green

PINCONFIGURATION

TOP VIEW



LGA-32 5*5

ABSOLUTE MAXIMUM RATING¹⁾

VIN.....	-0.3 to 20V
VIN(AC≤1μs)	24V
OUT.....	-0.3 to VIN
OUT (AC≤1ms)	-2V
VDD.....	-0.3V to 4.2V
All Other Pins.....	-0.3V to 4.2V
Junction Temperature ²³⁾	150°C
Lead Temperature	260°C
Storage Temperature.....	-65°C to +150°C

RECOMMENDED OPERATING CONDITIONS

VIN.....	4V to 16V
Maximum Continuous Output Current.....	0A to 50A
Operating Junction Temperature	-55 °C to 125°C
VDD Cap.....	min 1μF
CS/IMON Voltage Range.....	0V to 1.8V
CLREF Voltage Range.....	0.3V to 1.8V

THERMAL PERFORMANCE⁴⁾

	...	θ_{JA}	θ_{JC}
LGA-32 5*5.....	20	14.5°C/W	

Notes:

- 1) Exceeding these ratings may damage the device. These stress ratings do not imply function operation of the device at any other conditions beyond those indicated under RECOMMENDED OPERATING CONDITIONS.
- 2) The HQEF5016 includes thermal protection that is intended to protect the device in overload conditions. Continuous operation over the specified absolute maximum operating junction temperature may damage the device.
- 3) The device is not guaranteed to function outside of its operating conditions.
- 4) Measured on JESD51-7, 4-layer PCB.

ELECTRICAL CHARACTERISTICS

$V_{IN}=12V$, $V_{ON}=3.3V$, $C_{VDD}=1\mu F$, $C_{SS}=100nF$, $R_{CS}=R_{IMON}=2k\Omega$, $R_{CLREF}=120k\Omega$, $T_J=25^\circ C$, unless otherwise stated.						
ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNITS
VIN						
Quiescent current	I_Q	$V_{ON}=3.3V$, no load		1.5	2.1	mA
Shutdown current	I_{SHDN}	Fault latch off		1.3	1.8	mA
		$V_{ON}=0V$, $V_{IN}=16V$		1.1	1.6	mA
V_{IN} under-voltage lockout rising threshold	$V_{IN_UVLO_R}$		3.05	3.2	3.35	V
V_{IN} under-voltage lockout falling threshold	$V_{IN_UVLO_F}$		2.65	2.8	2.95	V
V_{IN} over-voltage protection(rising)	$V_{IN_OVP_R}$	V_{IN} rising, $T_J = -40^\circ C \sim 125^\circ C$	17.3	18.1	19.0	V
V_{IN} over-voltage protection(falling)	$V_{IN_OVP_F}$	V_{IN} falling, $T_J = -40^\circ C \sim 125^\circ C$	16.0	16.8	17.6	V
VDD Regulator						
VDD regulator output voltage	VDD	$I_{VDD}=0mA$	3.05	3.3	3.55	V
VDD regulator load capability	I_{VDD}		15		30	mA
VDD current limit	I_{DD_CL}		30	50		mA
VDD under-voltage lockout rising threshold	V_{DDUVLO_R}		2.55	2.75	2.85	V
VDD under-voltage lockout falling threshold	V_{DDUVLO_F}		2.15	2.3	2.45	V
VDD under-voltage lockout hysteresis	V_{DDUVLO_HYS}			450		mV
Power MOSFET						
On-Resistance	$R_{DS(ON)}$	$T_J=25^\circ C$		1	1.25	m Ω
		$T_J=125^\circ C$		1.35	1.625	m Ω
Off-state leakage current	I_{OFF}	$V_{IN}=16V$, $V_{ON}=0V$, $V_{OUT}=0V$ $T_J=25^\circ C$			2	μA
ON						
BIAS current	I_{ON}	$T_J = -40^\circ C \sim 125^\circ C$	3	4	5	μA
Turn on delay time	t_{ON}	Refer to Timing Waveforms	1.2	2	2.9	ms
Turn off delay time ⁵⁾	t_{OFF}	Refer to Timing Waveforms		2		μs

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ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNITS
Switch on threshold	V _{SWON}		1.3	1.4	1.5	V
Switch off threshold/Pull down upper threshold	V _{SWOFF}		1.1	1.2	1.3	V
Pull down lower Threshold	V _{PDOFF}		0.7	0.8	0.9	V
Pull down delay	t _{PD}	Output pull down delay timer		2		ms
Exit pull down mode delay ⁵⁾	t _{EPD}			2		μs
Pull down resistor	R _{PD}			540		Ω
IMON						
IMON sense gain	IMON _{GAIN}	10A≤I _{OUT} ≤20A	9.75	10	10.25	μA/A
		20A<I _{OUT} ≤30A	9.8	10	10.15	
		30A<I _{OUT} ≤40A	9.85	10	10.15	
		40A<I _{OUT} ≤50A	9.85	10	10.15	
CS						
CS sense gain	CS _{GAIN}	10A≤I _{OUT} ≤20A	9.7	10	10.3	μA/A
		20A<I _{OUT} ≤50A	9.8	10	10.2	
CURRENT LIMIT & CLREF						
Bias CLREF current	I _{CLREF}	T _J = -40°C~125°C	9.3	10	10.5	μA
CLREF internal current limit clamp voltage	V _{CLREF_CLAMP}	V _{CLREF_LO} at V _{OUT} <90%V _{IN} , T _J =25°C	580	635	690	mV
		V _{CLREF_LO} at V _{OUT} <90%V _{IN} , T _J =125°C	410	460	530	mV
		V _{CLREF_HI} at V _{OUT} >90%V _{IN}	1.9	2	2.1	V
CLREF operating voltage ⁵⁾	V _{CLREF}		0.3		1.8	V
CLREF over-current protection time	t _{CL_REG}	V _{OUT} >90%V _{IN} , V _{CLREF} >0.3V		312		μs
Short-circuit start-up protection timer	t _{SC_TIMER}	V _{OUT} <1/8*V _{IN} , Power MOSFET current is regulated by V _{CLREF_LO} ,		2	3.2	ms
Short-Circuit Current Limit Reference (SCREF)						
Internal SCREF current	I _{SCREF}	T _J =25°C	9.45	10	10.55	μA
		T _J = -40°C~125°C	9	10	11	μA
SCREF voltage range of short-circuit current limit	V _{SC_LIMIT}	SC limit=40A			0.16	V
		SC limit=60A	0.24	0.3	0.36	V
		SC limit=80A	0.48	0.6	0.72	V
		SC limit=120A	0.94	1.2	1.44	V

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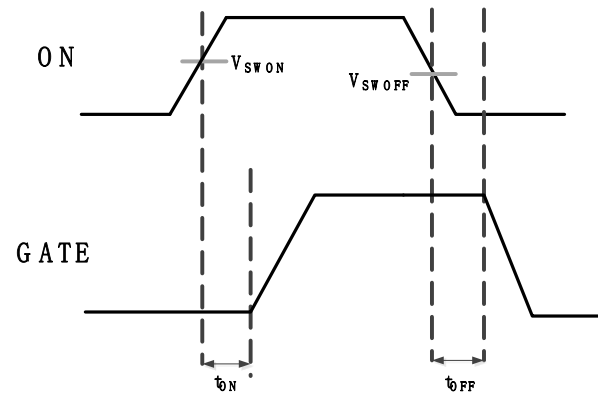
ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNITS
		SC limit=100A	1.6			V
Short-circuit current limit accuracy ⁵⁾	I _{SC_LIMIT}	V _{SCREF} > 1.6V		100		A
Response time ⁵⁾	t _{SC}	From I _{OUT} >I _{SC_LIMIT} until gate pull low to V _{OUT}		200		ns
VTEMP						
Gain ⁵⁾	A _{VTEMP}	Sensing rang 0°C to 125°C		8.7		mV/ °C
Bias Voltage ⁵⁾	V _{VTEMP25}	T _J =25°C		370		mV
Load capability	R _{VTEMP}	T _J =25°C	600			μA
Pull down current	I _{VTEMP}	T _J =25°C		45		μA
SS						
Bias current	I _{SS}	V _{IN} =12V, I _{SS} dependent V _{IN} , R _{SS} =0.8MΩ, T _J = -40°C~125°C	13	15	17	μA
SS pulldown voltage	V _{OL_SS}	Sink 0.1mA current		30		mV
GOK						
Output low Voltage	V _{OL_GOK}	I _{GOK} =1mA			0.1	V
Leakage current	I _{LKG_GOK}	V _{GOK} =3.3V			1	μA
GOK internal pull up resistor	R _{GOK_INT}			1.8		MΩ
GOK comparator low threshold	V _{GOK_TH}	T _J = -40°C~125°C	1.3	1.4	1.5	V
GOK comparator hysteresis	V _{GOK_HYS}			200		mV
GOK fault delay time	g _{GOK_dl}	T _J = -40°C~125°C	1.5		4	μs
D_OC						
Output low voltage	V _{OL_DOC}	I _{D_OC} =1mA			0.1	V
Leakage current	I _{LKG_DOC}	V _{D_OC} =3.3V			1	μA
Over-current threshold	V _{OC_TH}	V _{OUT} ≥90%*V _{IN}	82	86	90	%V _{CLREF}
D_OC delay time(rising)	t _{DOC_R}	V _{CS} <V _{OC_TH} until D_OC rising		1		μs
D_OC delay time(falling)	t _{DOC_F}	V _{CS} >V _{OC_TH} until D_OC falling		1		μs
Thermal Shutdown						
Over-temperature shutdown ⁵⁾	t _{STD}			140		°C
Thermal shutdown hysteresis ⁵⁾	t _{HYS}			20		°C
FAULT Detection						

$V_{IN}=12V$, $V_{ON}=3.3V$, $C_{VDD}=1\mu F$, $C_{SS}=100nF$, $R_{CS}=R_{IMON}=2k\Omega$, $R_{CLREF}=120k\Omega$, $T_J=25^\circ C$, unless otherwise stated.

ITEM	SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNITS
V_{DS} short threshold	V_{DS_TH}	Start-up postponed if $V_{OUT}>V_{DS_TH}$ at $V_{ON}>V_{SWON}$ transition	85	90	95	% V_{IN}
V_{DS} short OK threshold	V_{DS_OK}	Start-up resumed if $V_{OUT}<V_{DS_TH}$ any time after postponed	65	70	75	% V_{IN}
V_{GS} short threshold ⁵⁾	V_{GS_TH}	$V_{GS}<V_{GS_TH}$		2.2		V
V_{OUT} low threshold	V_{OUTL_TH}			90		% V_{IN}
Gate fault timer ⁵⁾	t_{GATE_FLT}	Time from $V_{GS}<V_{GS_TH}$ after ss-end		200		ms
Start-up timer failsafe ⁵⁾	t_{SS_END}			200		ms
Fault Type Indication Output (FLT_TYPE)						
FLT_TYPE voltage at over-current protection	$V_{FLT_TYPE_OC}$	Over-current fault at normal and soft-start	1.14	1.2	1.26	V
FLT_TYPE voltage at short-circuit protection	$V_{FLT_TYPE_SC}$	Short-circuit fault	1.40	1.5	1.59	V
FLT_TYPE voltage at over temperature, V_{IN} OVP	$V_{FLT_TYPE_OT}$	Over temperature and OVP fault	0.84	0.9	0.95	V
FLT_TYPE voltage at MOSFET DS and GD short	$V_{FLT_TYPE_DSS}$	MOSFET not fully on (GS/DS short, charge pump fault or Gate Leakage fault)	0.55	0.6	0.65	V
FLT_TYPE voltage at GOK Fault	$V_{FLT_TYPE_GSS}$	GOK (comparator) fault event	0.25	0.3	0.34	V
FLT_TYPE voltage at others	$V_{FLT_TYPE_other}$	FLT_TYPE voltage at others (normal operation)	0.05	0.1	0.15	V
MODE						
MODE sourcing current	I_{MODE}	$T_J=-40^\circ C\sim 125^\circ C$	9	10	11	μA
Hiccup with 500ms retry delay MODE window	V_{MODE1}	$R_{MODE}=GND$		0	0.16	V
Hiccup with 250ms retry delay window	V_{MODE2}	$R_{MODE}=30k\Omega$	0.24	0.3	0.35	V
Hiccup with 1s retry delay window	V_{MODE3}	$R_{MODE}=60k\Omega$	0.48	0.6	0.72	V
Latch off MODE window	V_{MODE5}	Pull up to VDD or floating	1.6			V

Note:

5) Guaranteed by design.



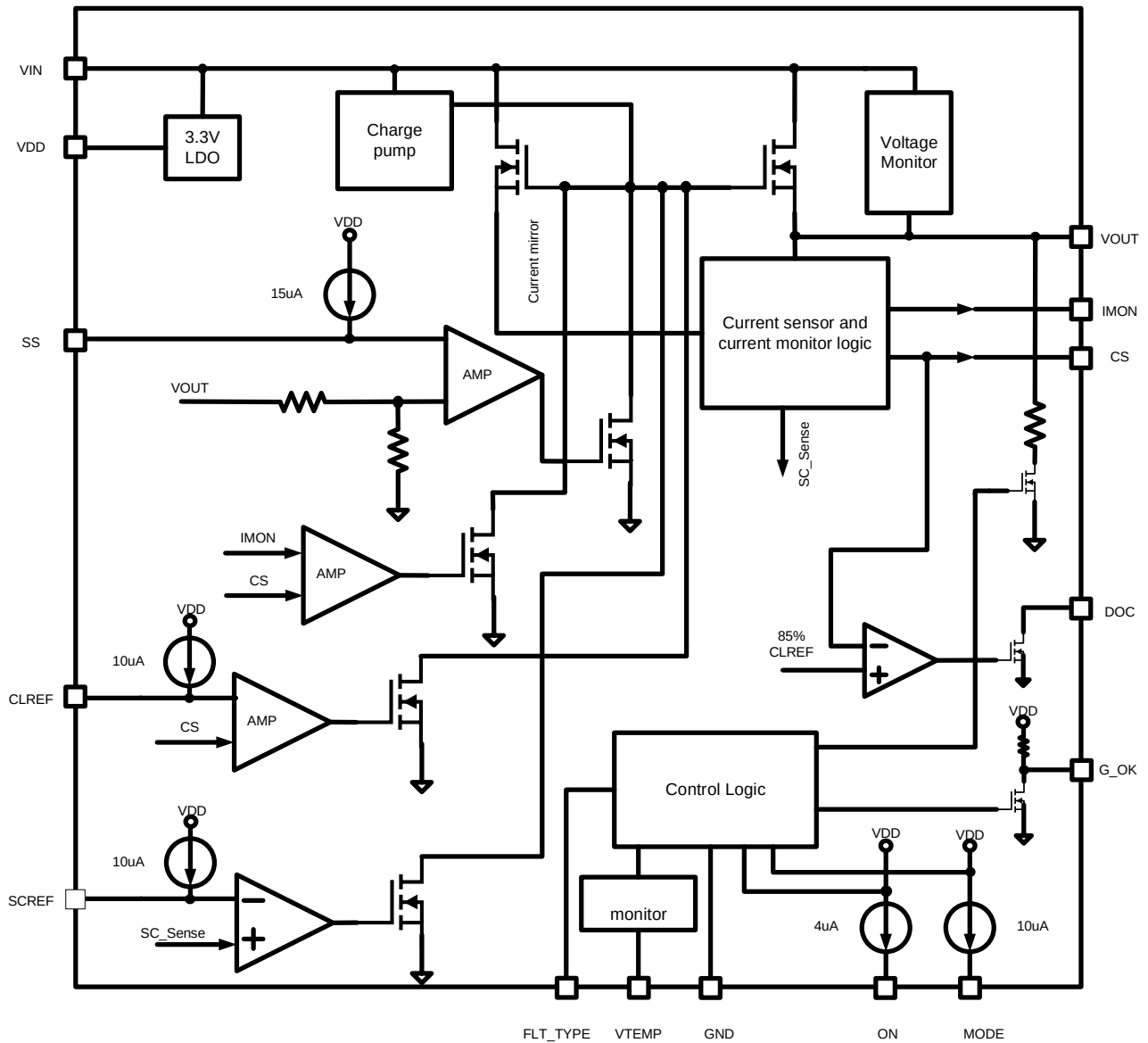
Timing Waveforms

PIN DESCRIPTION

HQEF5016	NAME	DESCRIPTION
PIN		
1	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
2	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
3	D_OC	Open drain over-current indicator. D_OC is pulled low when over-current event occurs. D_OC does not report current limit during soft-start.
4	ON	Active high. Drive ON pin high to enable power MOSFET and drive ON pin around 1V to turn on pull down resistance. Turn off MOSFET by drive ON pin low.
5	GOK	Active high. GOK is pulled low when fault condition occurs.
6	FLT_TYPE	Fault type indication output. FLT_TYPE indicate a voltage level when an associated fault occurs.
7	NC	No connection. Leave it float.
8	MODE	Mode selection. Connect a resistor from MODE to GND to select the operation mode including: latch off or hiccup operation, hiccup retry delay time.
9	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
10	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
11	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
12	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
13	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
14	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
15	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
16	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.
17	SCREF	Short-circuit current limit. HQEF5016sample SCREF voltage during the insertion delay time after VDD passing UVLO, and select short-circuit current limit as one of 40/60/80/100/120A.
18	VTEMP	Analog temperature sense output. Place a $\leq 1\text{nF}$ capacitor.
19	SS	Soft-start time set pin. An external capacitor need to be connected from this pin to GND to program soft-start time.
20	GND	Ground.
21	VDD	Internal 3.3V LDO output.
22	IMON	Analog current monitor output.
23	CS	Current sense output.
24	CLREF	Current limit reference pin. Current limit can be programmed by forcing this pin to different voltage.
25	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
26	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
27	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
28	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
29	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
30	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.

31	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
32	VOUT	Output pin of E-fuse. Connected to source of power MOSFET.
Exposed Pad	VIN	Input pin of E-fuse. Connected to drain of power MOSFET.

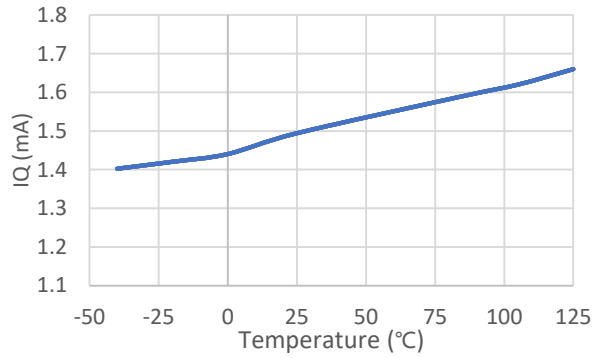
BLOCK DIAGRAM



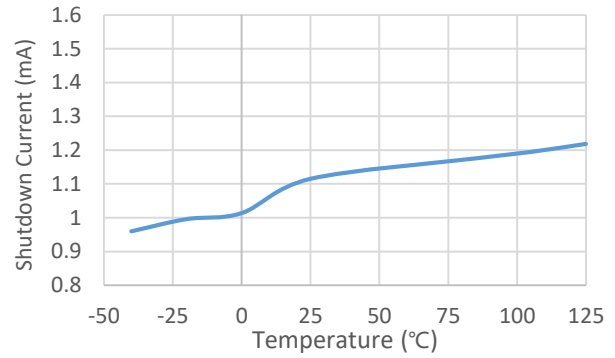
TYPICAL PERFORMANCE CHARACTERISTICS

$V_{IN} = 12V$, $T_A = +25^\circ C$, $R_{CS} = R_{IMON} = 2k\Omega$, unless otherwise noted

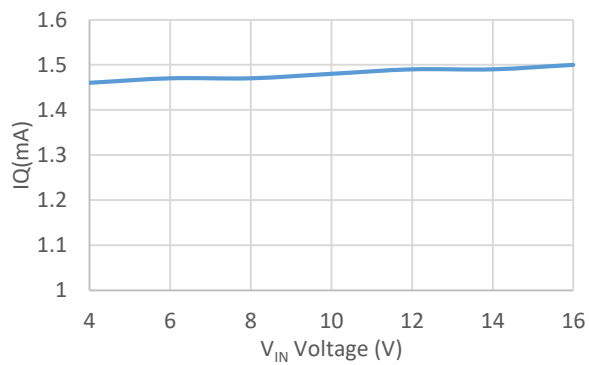
I_Q vs. Temperature



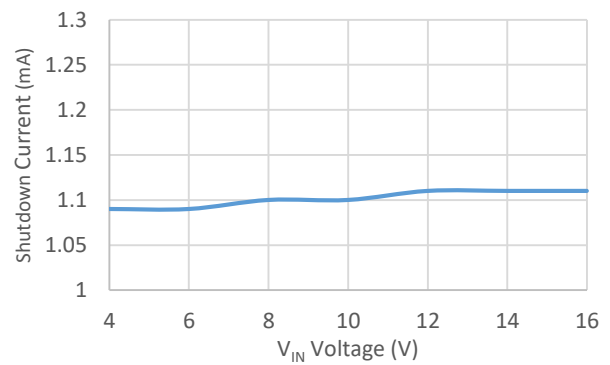
Shutdown Current vs. Temperature



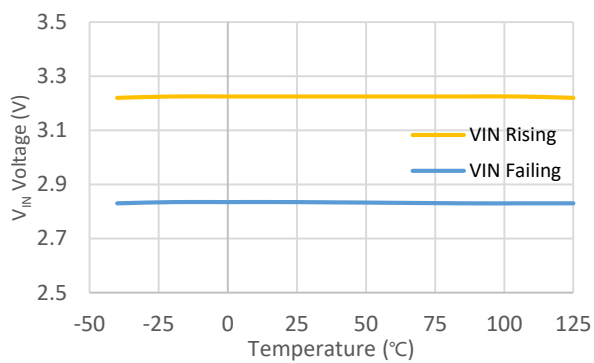
I_Q vs. V_{IN} Voltage



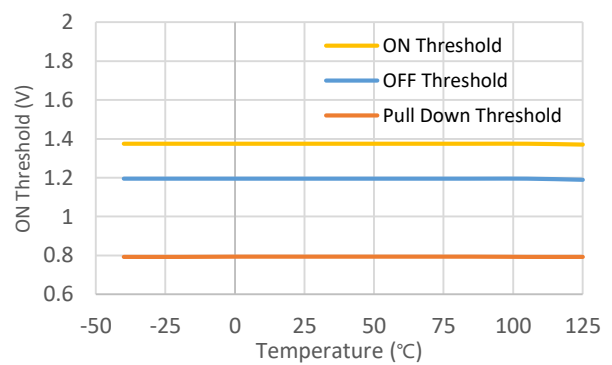
Shutdown Current vs. V_{IN} Voltage



V_{IN_UVLO} vs. Temperature



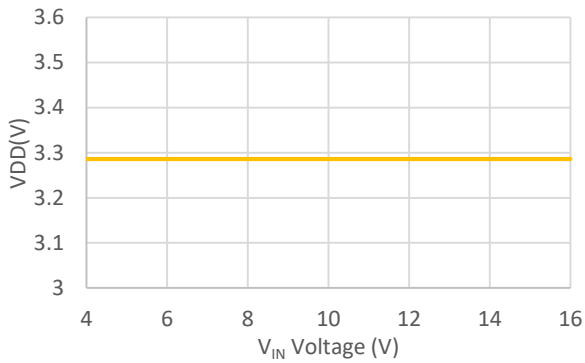
ON Threshold vs. Temperature



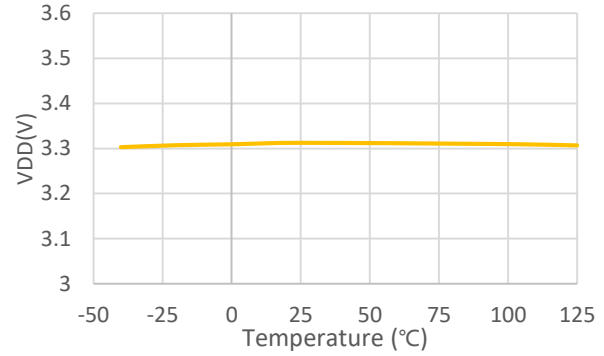
TYPICAL PERFORMANCE CHARACTERISTICS(continued)

$V_{IN} = 12V$, $T_A = +25^\circ C$, $R_{CS} = R_{IMON} = 2k\Omega$, unless otherwise noted

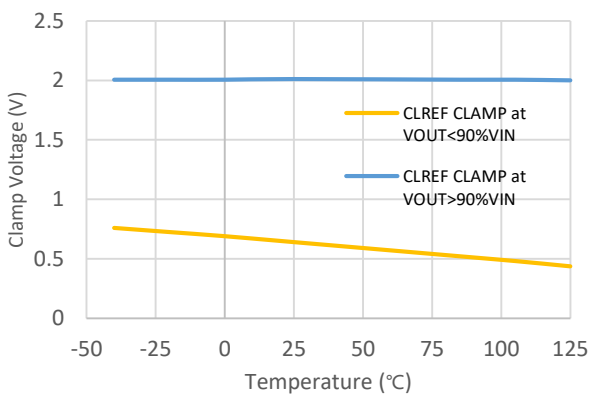
VDD Voltage vs. V_{IN} Voltage



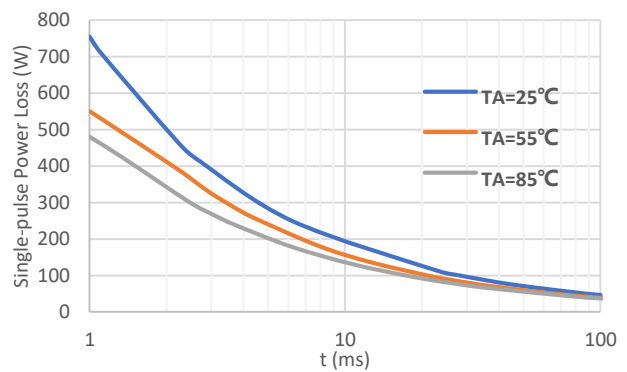
VDD Voltage vs. Temperature



CLREF Clamp Voltage vs. Temperature

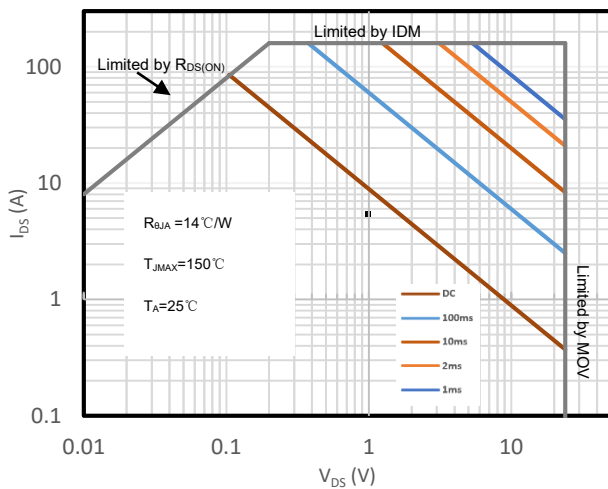


Single-Pulse Power Loss vs. Thermal Shutdown Time

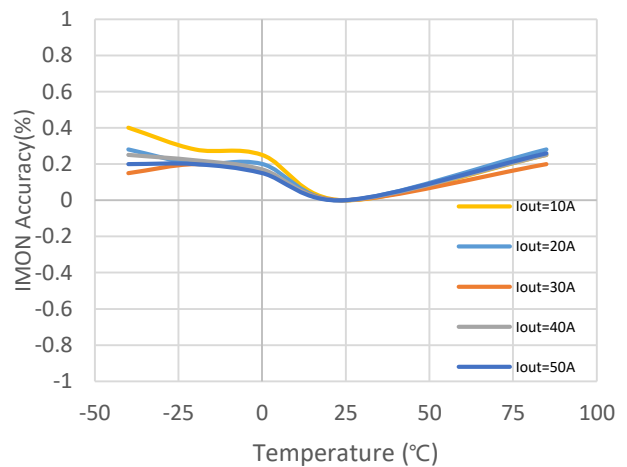


Safe Operation Area (SOA)

Tested on the EVH7160A(10cmx14cm), 4-layer PCB, 2oz per layer



IMON Accuracy vs. Temperature

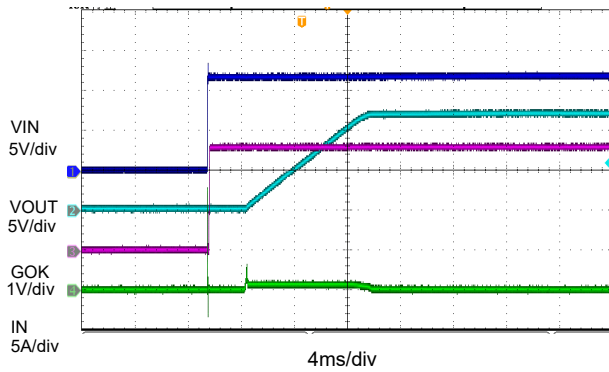


TYPICAL PERFORMANCE CHARACTERISTICS(continued)

$V_{IN} = 12V$, $T_A = +25^{\circ}C$, $R_{CS} = R_{IMON} = 2k\Omega$, $C_{out} = 470\mu F$, $C_{SS} = 100nF$, unless otherwise noted

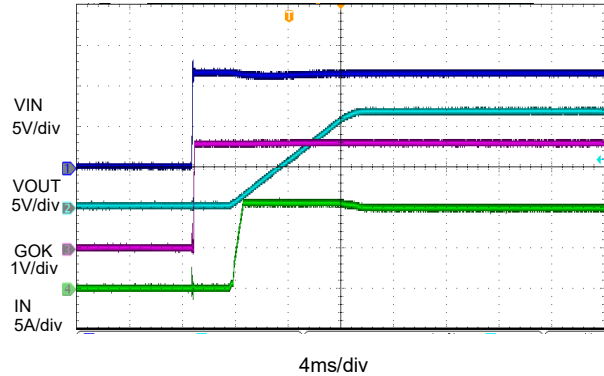
VIN Hot Plug

$I_{OUT} = 0A$



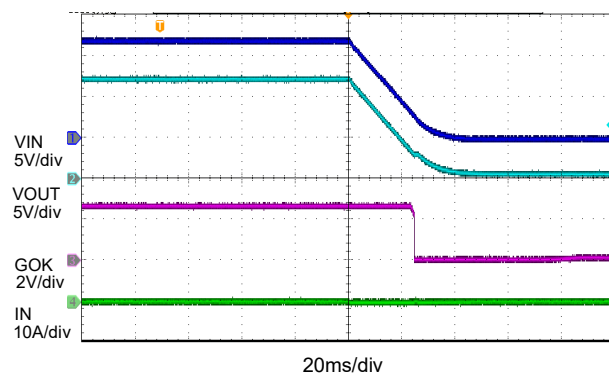
VIN Hot Plug

$I_{OUT} = 10A$



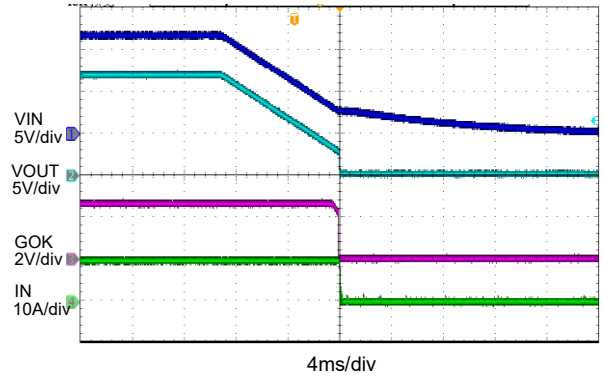
Shutdown through VIN

$I_{OUT} = 0A$



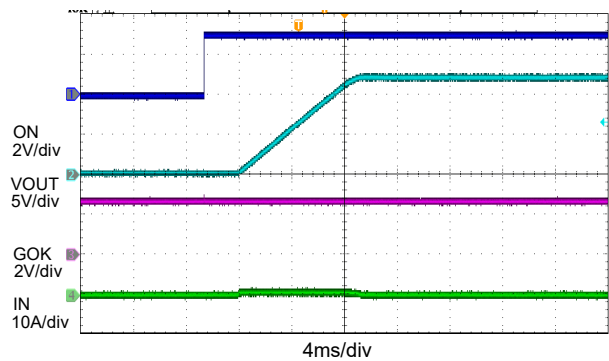
Shutdown through VIN

$I_{OUT} = 10A$



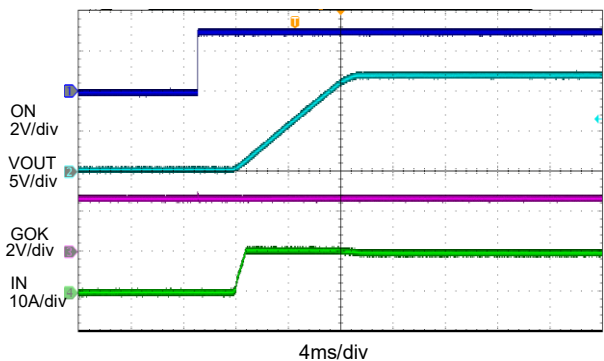
Start-Up through ON

$I_{OUT} = 0A$



Start-Up through ON

$I_{OUT} = 10A$

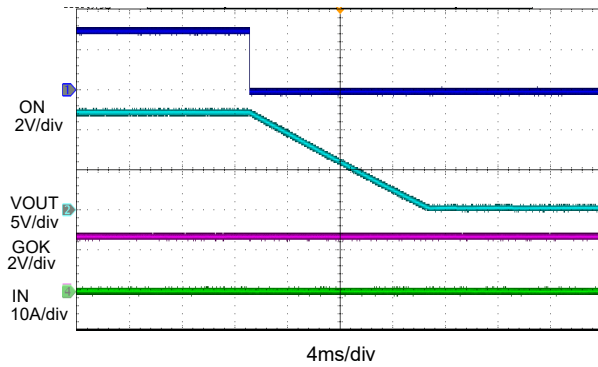


TYPICAL PERFORMANCE CHARACTERISTICS(continued)

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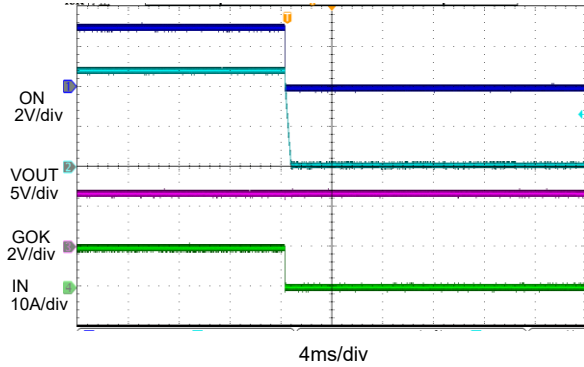
Shutdown through ON

$I_{OUT}=0A$



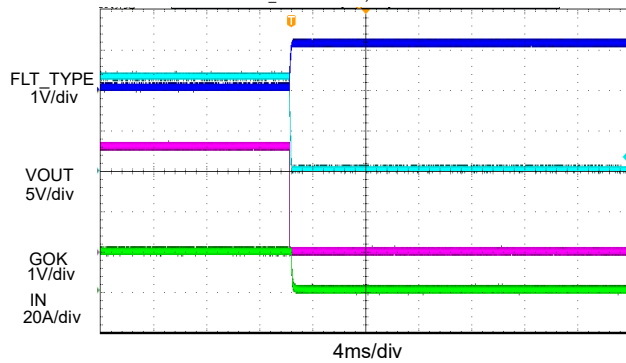
Shutdown through ON

$I_{OUT}=10A$



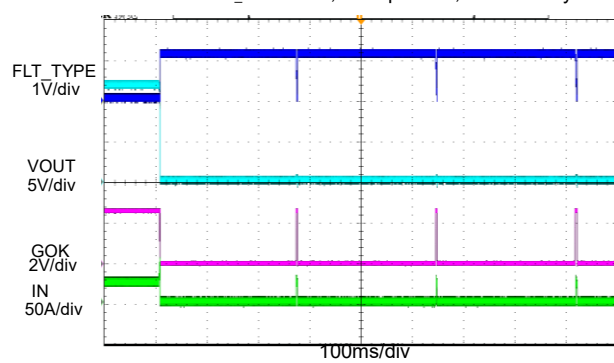
Over-Current Protection

$I_{OC_normal}=50A$, latch mode



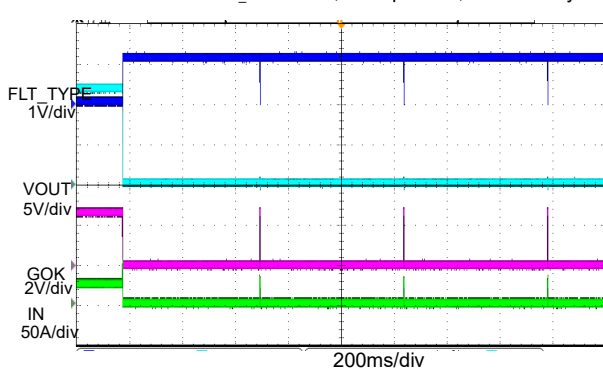
Over-Current Protection

$I_{OC_normal}=50A$, hiccup mode, 250ms retry



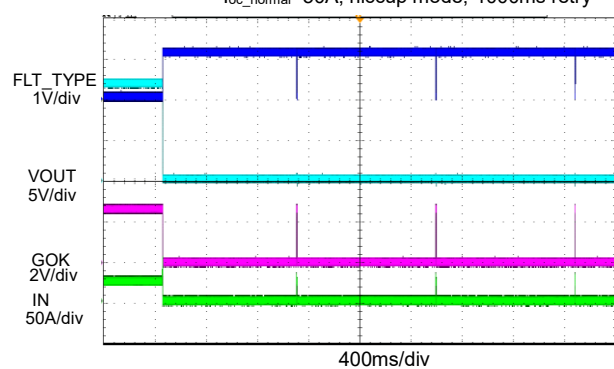
Over-Current Protection

$I_{OC_normal}=50A$, hiccup mode, 500ms retry



Over-Current Protection

$I_{OC_normal}=50A$, hiccup mode, 1000ms retry

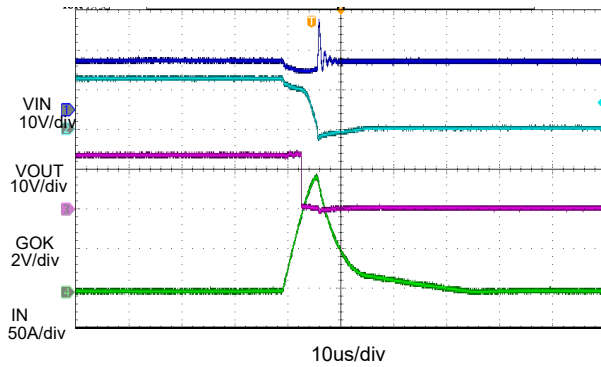


TYPICAL PERFORMANCE CHARACTERISTICS(continued)

$V_{IN} = 12V$, $T_A = +25^{\circ}C$, $R_{CS} = R_{IMON} = 2k\Omega$, $C_{out} = 470\mu F$, $C_{SS} = 100nF$, unless otherwise noted

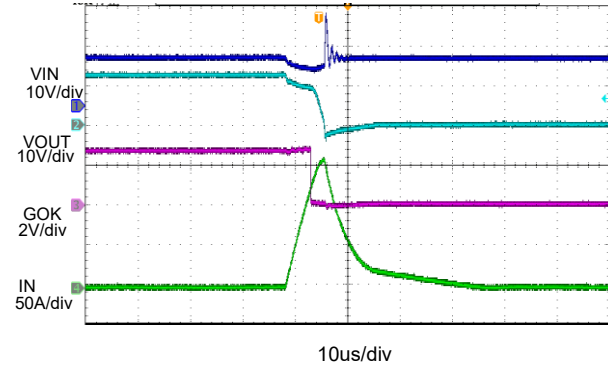
Short-Circuit Protection

$I_{SC_LIMIT}=80A$, latch mode,



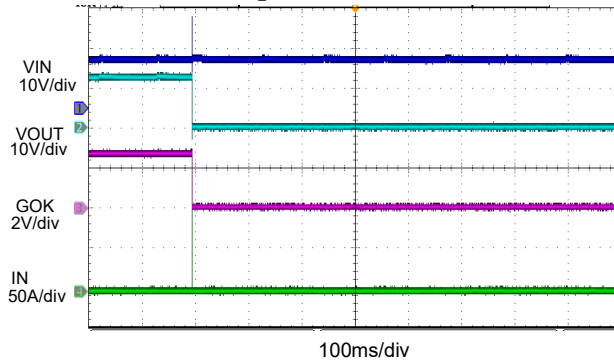
Short-Circuit Protection

$I_{SC_LIMIT}=120A$, latch mode,



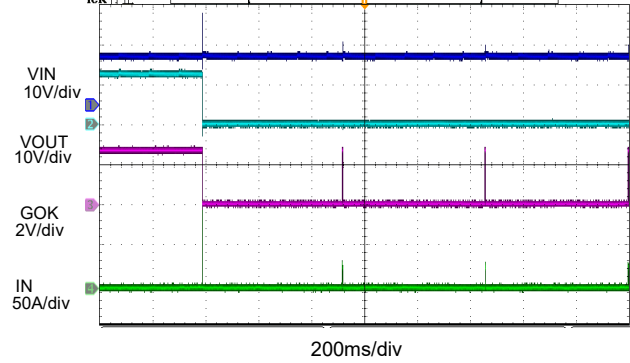
Short-Circuit Protection

$I_{SC_LIMIT}=120A$, latch mode,



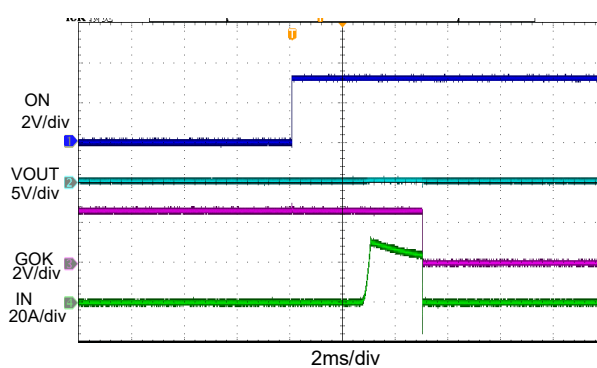
Short-Circuit Protection

$I_{SC_LIMIT}=120A$, hiccup mode, 500ms retry



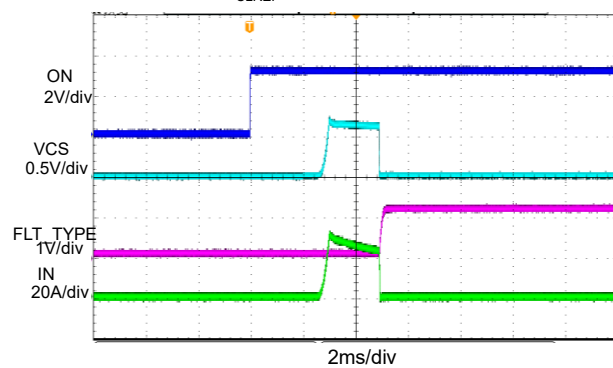
Short-Circuit Start-Up

$V_{CLREF}=1.2V$



Short-Circuit Start-Up

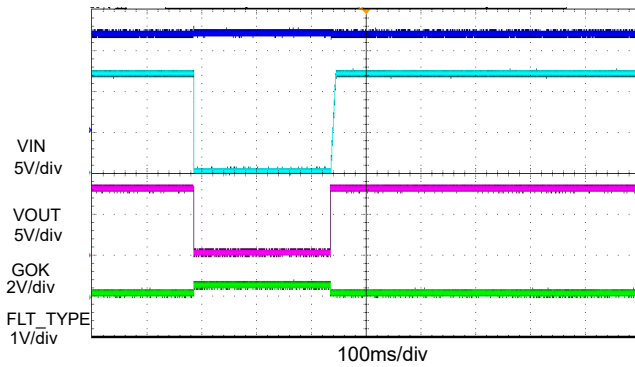
$V_{CLREF}=1.2V$



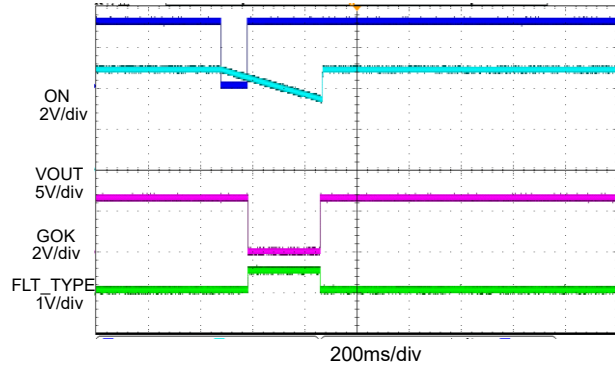
TYPICAL PERFORMANCE CHARACTERISTICS(continued)

$V_{IN} = 12V$, $T_A = +25^{\circ}C$, $R_{CS} = R_{IMON} = 2k\Omega$, $C_{out} = 470\mu F$, $C_{SS} = 100nF$, unless otherwise noted

GOK Fault Protection

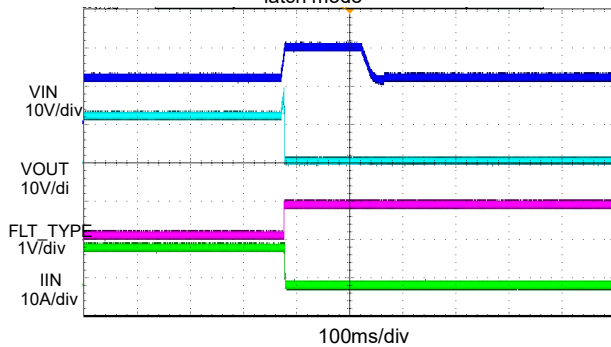


DS Short Protection



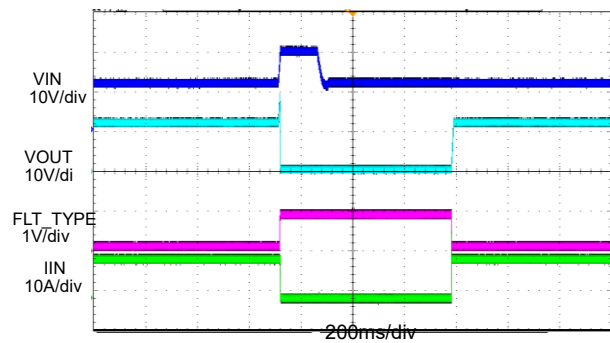
OVP Protection

latch mode



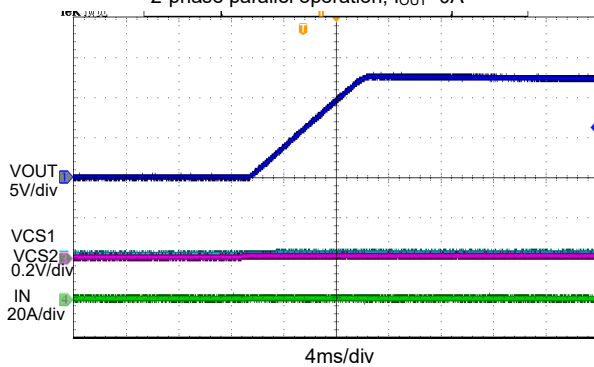
OVP Protection

hiccup mode, 500ms retry



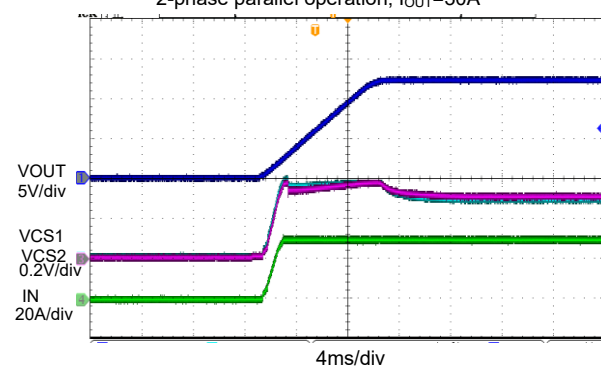
Current Balance during Start-Up

2-phase parallel operation, $I_{OUT}=0A$



Current Balance during Start-Up

2-phase parallel operation, $I_{OUT}=30A$



FUNCTION DESCRIPTION

The HQEF5016 is an electronic fuse that provides inrush current limit protection thereby allowing a board to be safely inserted and removed from a live backplane. It integrates a low $R_{DS(ON)}$ MOSFET that provides accurate current, temperature monitor function and eliminates the need for an external current-sense power resistor, power MOSFET, and thermal sense device.

Current Limit at Start-up

During start-up operation, the start up current is limited by the current limit reference voltage and the CS sense voltage. The CS voltage is compared with the reference voltage to adjust the gate voltage of the power MOSFET through the op amp, to achieve the purpose of limiting the starting current.

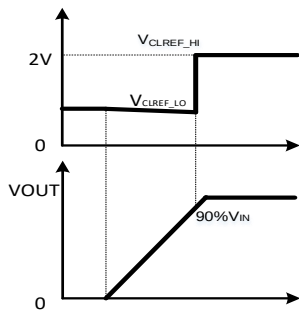


Figure 1: CLREF Clamp According to V_{out}

In the start-up phase, $V_{OUT} < 90\%V_{IN}$, the reference voltage can be obtained by external setting or internal limit, usually this is the internal voltage clamping reference value V_{CLREF_LO} (typical value 635mV) to ensure that the current is limited during the start-up process, on the power MOSFET the loss will not be too great. When the output voltage climbs close to V_{IN} , $V_{OUT} > 90\%V_{IN}$, the reference voltage V_{CLREF_HI} will climb to a large reference value of 2V. If the

voltage set via the external CLREF is less than the internal clamp voltage (V_{CLREF_CLAMP}), the actual voltage reference is determined by the CLREF voltage.

The starting process current limit can refer to the following figure 1.

In the start-up phase, when the CS voltage exceeds the reference voltage, if $V_{OUT} < 1/8V_{IN}$, the power MOSFET current will be limited by V_{CLREF_LO} regulation for 2.1ms, if the fault current persists after 2.1ms, the power MOSFET will be turned off and restart or latch. If $1/8V_{IN} < V_{OUT} < 90\%V_{IN}$, the current of the power MOSFET will be limited by V_{CLREF_LO} regulation for 0.312ms. If the fault current persists after 0.312ms, the power MOSFET will be turned off and restarted or latched. The current limit during this time is referred to the schematic diagram of Figure 2.

$$I_{LIMIT_Satrt} = \frac{\min(V_{CLREF_LO}, V_{CLREF})}{R_{CS} * CS_{GAIN}}$$

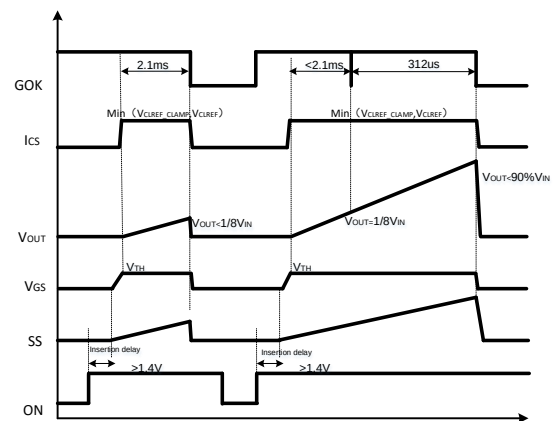


Figure 2: Failed Start-up with Output Current Limit

During start-up, if $V_{OUT} > 1/8V_{IN}$, but gate to source voltage is lower than the charge pump voltage after $SS > 2.5V$, the power MOSFET is turned off after 200ms delay time. It is recognized as a faulty condition that the power

MOSFET can't be fully enhanced, and GOK is pulled low.

The chip possesses robust protection capabilities, capable of handling short-circuit scenarios even under extremely adverse conditions. For instance, situations like a significant parasitic inductance in the output short-circuit loop or an incomplete short-circuit connection at the output could potentially cause minor current oscillations near the current limit, leading to detection inaccuracies. Nevertheless, the chip still employs sensitive temperature sampling to detect over-temperature conditions and provide protection.

Circuit Breaker at Normal Operation

During the normal state, the HQEF5016 responds to the output overcurrent condition after a fixed protection counting time. This circuit breaker protection allows allowing the system to withstand transient peak currents of the load without causing fluctuations in electronic fuse.

When the output voltage has ramped up close to V_{IN} , and V_{OUT} exceeds 90% of V_{IN} , the internal current limit reference clamp voltage is released to 2V, and the power MOSFET gate voltage is close to the internal charge pump voltage. Once the HQEF5016 detects that start-up has finished, the part operates normally.

During normal operation, When the CS voltage exceeds the reference voltage (the minimum value of the external setting of CLREF and the internal reference V_{CLREF_CLAMP}), that is the load current exceeds the programmed current limit threshold (I_{OC_normal}), but stays lower than the short-circuit threshold (I_{SC_LIMIT}), the over-current counter starts to count. If the fault current persists after 312 μ s, the chip will turn off the power MOSFET and perform a restart or latch state. If the V_{CS} falls below the reference voltage within the 312 μ s fault response time, the chip operates normally.

$$I_{OC_normal1} = \frac{\min(V_{CLREF_CLAMP}, V_{CLREF})}{R_{CS} * CS_{GAIN}}$$

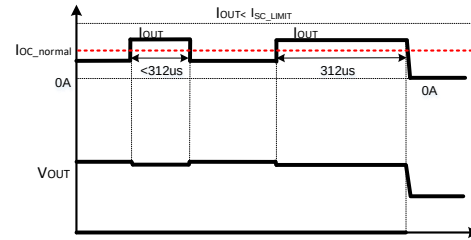


Figure 3: Over-Current Protect at Normal Operation

Power Up Sequence

For hot-swap applications, while the input voltage rises immediately, the power MOSFET gate voltage should always be pulled low during the V_{IN} plug-in with high slew rate. If the HQEF5016 works in hot swap application, an external capacitor (C_{ON}) can be connected from ON to ground for insert delay start-up (see Figure 4). The internal 4 μ A current source charges the capacitor when V_{DD} exceeds the V_{DDUVLO_R} . ON can also be pulled up externally to the V_{DD} voltage. A 10 μ A CLREF current source determines the over-current limit level through R_{CLREF} . A 10 μ A SCREF current source determines the short-circuit current limit level through R_{SCREF} .

Once the ON voltage exceeds 1.4V and the insertion delay time ends, the power MOSFET is charged up by the internal charge pump. The power MOSFET turns on when V_{GS} reaches on threshold of the power MOSFET (V_{TH}), and then the output voltage rises. The current limit reference voltage is clamped internally to V_{CLREF_LO} when the external CLREF voltage is higher than the internal V_{CLREF_LO} clamp. If the external CLREF voltage is below the clamp voltage, the actual current limit reference is determined by the CLREF voltage.

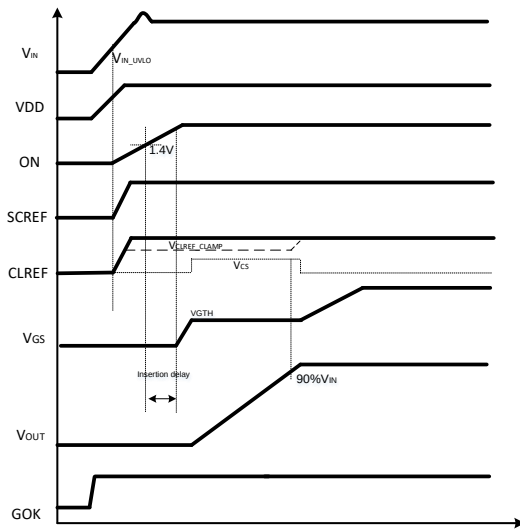


Figure 4: Start-up Timing

The SCREF voltage is sampled during the first insertion delay time after VDD passes UVLO threshold and determine the short-circuit current limit. The short-circuit current limit is held after the insertion delay time ends.

Soft-start (SS)

A capacitor connected to SS determines the soft-start time. When ON/PD is pulsed high and the insertion delay time ends, a constant current source charges up the SS voltage. The output voltage follows the SS voltage.

In parallel applications, the SS pins of all fuses should be shorted together to one shared SS capacitor. Internal soft-start load balancing circuitry will ensure the soft-start current is shared between paralleled devices.

The soft-start capacitor value can be calculated by below equation:

$$C_{SS} = 9 * t_{SS} / R_{SS}$$

Where t_{SS} is the soft-start time(ms), C_{SS} is the soft-start capacitor(nF), and R_{SS} is 0.8MΩ.

For example, a 100nF capacitor sets t_{SS} to 8.9ms. Float the SS pin to generate a fast ramp-up

voltage, the starts up time is 4.5ms.

ON/PD Control

ON/PD can be used for on/off of the power MOSFET and the pull down mode of the output voltage. When $V_{ON} > V_{SWON}$ the device is enabled. When V_{ON} lower than V_{SWOFF} , the device is disabled.

When the ON/PD voltage is set to between 1.2V to 0.8V, the device works in pull down mode. In pull down mode, when the power MOSFET is turned off, an integrated 540Ω attached to the output discharges the output voltage after a fixed delay time (typical 2ms). If the ON/PD is pulled low directly, pull down mode is disabled, output voltage discharge only can rely on the external load.

The ON/PD pin has an internal pull up current, if this pin is floating, the device goes to the ON state. This pull up current can be used to charge an external capacitor, in this way, the initial delay time was extended.

DOC

The HQUEF5016 has over-current indicator output. The DOC pin is the open drain of a MOSFET. Connect to a voltage source (such as 3.3V) through a 1kΩ to 100kΩ resistor, active low output that reports overcurrent warnings while V_{OUT} exceeds 90% of V_{IN} . When V_{CS} exceeds 85% of V_{CLREF} , the DOC signal becomes low.

Current Limit Reference (CLREF)

The CLREF pin voltage determines the current limit regulation point. The CLREF voltage can be applied by an external source, or developed across a programming resistor ground by the CLREF bias current. The recommended range of CLREF voltage is 0.3V to 1.8V.

$$V_{CLREF} = R_{CLREF} * I_{CLREF}$$

Current Sense Output (CS)

The CS pin is both an input and an output. The CS pin sources a current that is CS_{GAIN} times the V_{OUT} current. This produces a voltage on the CS pin that is the product of the CS pin current and an external CS pin resistance to ground. The gain of the current sense is $10\mu A/A$. Connect a resistor (R_{CS}) to ground set the current sense voltage V_{CS} . This voltage determines the D_{OC} indicator trip point and the current regulation point, via its interaction with the voltage on $CLREF$ pin.

During normal operation, if the voltage on the V_{CS} pin is higher than V_{CLREF} , it enters the current protection counting state. After exceeding the overcurrent protection time, the gate voltage of the power MOSFET is pulled low, and the output limited current is calculated by the following formula.

$$I_{OUT} = V_{CLREF} / (R_{CS} * CS_{GAIN})$$

During soft-start, the current limit reference voltage is clamped according to the below table 1. The CS_{GAIN} at start-up state is slightly higher than normal state.

Table1.The Current Limit Threshold

	V_{CL_TH}
$V_{OUT} < 90\% * V_{IN}$	V_{CLREF_LO} or V_{CLREF} (whichever is lower)
$V_{OUT} > 90\% * V_{IN}$	V_{CLREF_HI} or V_{CLREF} (whichever is lower)

If an over-current condition exists anytime for a continuous duration t_{CL_REG} , then devices into latch off or hiccup mode (set by mode pin). During start-up, if V_{OUT} is below $1/8 * V_{IN}$ and the power MOSFET current is regulated by V_{CLREF_LO} 2.1ms, the power MOSFET is turned off as faulty condition, and GOK is pulled low, then devices into latch off or hiccup mode.

Current Monitor Output (IMON)

The gain of the current monitor is $10\mu A/A$. Connect a resistor (R_{IMON}) to ground set the gain

of the output voltage. The output voltage can be used by the controller to accurately monitor the output current. To keep have one good linearly proportional, the IMON voltage range of 0V to 1.8V is required.

It is important to note that the voltage setting on the IMON pin should be higher than CS. To achieve balanced current characteristics during start-up, It is recommended R_{IMON} should be $\geq R_{CS}$. It is recommended to place a capacitor of no more than 100nF in parallel or without adding a capacitor between IMON and ground.

Temperature Sense Output (VTEMP)

V_{TEMP} is a voltage output proportional to the junction temperature. The voltage is $8.7mV/^{\circ}C$ with a 152.5mV offset, and can be calculated with equation:

$$V_{TEMP} = T_j * 8.7mv + 152.5mV$$

For example, if the junction temperature is $25^{\circ}C$, V_{TEMP} is about 0.37V. The total temperature sense range is $-20^{\circ}C$ to $+140^{\circ}C$.

V_{TEMP} is a voltage output proportional to the device temperature and has a fixed offset voltage. In scenarios where multiple devices are used in parallel, connecting multiple V_{TEMP} pin together will result in V_{TEMP} outputting the voltage generated by the hottest HQUEF5016 among them. It is recommended to connect a $\leq 1nF$ capacitor from the V_{TEMP} pin to ground.

VDD Regulator

The HQUEF5016 has an internal 3.3V linear sub-regulator. This VDD can be driven with an external 3.3V to reduce sub-regulator loss from V_{IN} . This auxiliary output supply is current-limited to I_{DD_CL} . A ceramic capacitor in the range of 1uF to 10uF must be placed between the VDD and GND pins, as close to the HQUEF5016 as possible.

Short-circuit Protection (SCP)

If the load current increase rapidly due to a short circuit, the current may exceed the current limit threshold significantly before this device control loop can response. The device integrates a short-circuit protection, if the current reaches the I_{SC_LIMIT} (set by the SCREF pin), a fast turn off circuit is activated to turn off the power MOSFET. The total short-circuit response time is about 200ns.

SCREF

The HQUEF5016 provides external adjustable short-circuit protection threshold. Table 2 below shows the protection threshold.

Table 2. Short Protect Current Selection

V_{SC_LIMIT} (V)	SCREF pin connection R_{SCREF}	SC LIMIT (A)
<0.16	GND	40
0.24~0.36	30k Ω	60
0.48~0.72	60.4k Ω	80
0.94~1.44	120k Ω	120
>1.6	200k Ω	100

If the load current increases rapidly due to a short circuit, the current may exceed the current limit threshold significantly before the hot-swap control loop can respond. If the current reaches short-circuit current limit, a fast turn off circuit in the is activated to turn off the power MOSFET. The total short-circuit response time is about 200ns. The GOK signal asserts low when the power MOSFET current reaches the short-circuit current.

The HQUEF5016 samples the SCREF voltage during the first insertion delay time after VDD passes its UVLO threshold to determine the short-circuit current limit. The current limit is hold when insertion delay time ends. Change SCREF voltage after insertion delay time won't affect the short-circuit current anymore.

Thermal Protection

The HQUEF5016 integrates one on-die temperature sensing to monitor the junction temperature of the IC. The temperature information can be read from VTEMP. Also, the device has one thermal protection. When the junction temperature exceeds the threshold(140°C), the power MOSFET is turned off, then the power MOSFET latches/restarts.

In latch off mode, when the temperature drops below the recovery temperature, the chip will also be in a faulty state. Only by resetting VIN or ON pin can the chip clear the fault and initiate a normal start-up process again.

In hiccup mode, when the temperature drops below 120°C, the chip will initiate an automatic retry. After a t_{RETRY} time, the GOK signal is pulled high, and the chip begins a normal start-up process.

Under Voltage Lockout Protection(UVLO)

The HQUEF5016 has two under voltage lock-out protections: a 2.4V V_{DD_ULVO} and 2.9V V_{IN_UVLO} . When start-up, the device not work until only both VDD and V_{IN} exceed their respective rising UVLO thresholds. When shutdown, any one of VDD or V_{IN} lowers than their respective falling UVLO thresholds, the device will be off.

VIN Over-Voltage Protection (OVP)

The HQUEF5016 monitors V_{IN} to determine whether an over-voltage (OVP) fault has occurred. If V_{IN} exceeds 18.1V, then the power MOSFET turns off, and the GOK and SS pins are pulled low.

In latch off mode, when the V_{IN} voltage drops below the OVP recovery threshold, the chip will also be in a faulty state. Only by resetting VIN or ON pin can the chip clear the fault and initiate a normal start-up process again.

In hiccup mode, when the V_{IN} voltage drops

below 16.8V, the chip will initiate an automatic retry. After a t_{RETRY} time, the GOK signal is pulled high, and the chip begins a normal start-up process.

MODE

The HQUEF5016 provides both latch off and hiccup operation for faults protections of input over-voltage protection, output over-current protection, short-circuit protection, over temperature protection and power MOSFET GS/GD short protection. Configure MODE pin external resistor can set latch off or hiccup protection together with retry delay time.

Table 3 below shows the protection mode, retry delay time. For the Hiccup mode, after the retry delay time, the device will re-start through another one soft-start process. For the latch off mode, the device will not power on until the next power recycling.

When the device passes VDD and V_{IN} UVLO threshold, the MODE pin will be detected and once the setting is detected it can't change until the next VDD and VIN power recycling.

Table 3. Mode Selection

Mode pin connection	Mode	Retry delay time(t_{RETRY})
GND	Hiccup	500ms
30k Ω	Hiccup	250ms
60k Ω	Hiccup	1s
Pull up to VDD or floating	Latch off	

MOSFET Detection

The HQUEF5016 can detect the internal power MOSFET drain-to-source short before the power MOSFET is turned on. When the VDD voltage and ON voltage are higher than the rising threshold, HQUEF5016 starts a short-circuit detection. When HQUEF5016 detects that $V_{OUT} > 90\% V_{IN}$, it is judged that a DS short-circuit

has occurred, GOK is pulled low, and the FLT_TYPE voltage is indicated as 0.6V. When the short-circuit is removed, the HQUEF5016 detects that $V_{OUT} < 70\% V_{IN}$, the GOK signal is pulled high, the FLT_TYPE voltage indicates 0.1V, and the HQUEF5016 restarts the normal start-up process.

GATE to VIN short, If the device is disabled and time from V_{GS} short after Gate fault timer (200ms), then GOK is pulled low and the FLT_TYPE voltage indicates 0.6V. In hiccup mode, the device allowed to power up once V_{GS} short remove before Gate fault timer.

GATE leakage in start-up. If $V_{GS} < V_{GS_TH}$ at t_{SS_END} , then GOK is pulled low and power MOSFET latches/restarts.

GATE leakage in normal operation. If $V_{GS} < V_{GS_TH}$ for t_{GATE_FLT} time after the soft-start timer completes, then GOK is pulled low and device latches/restarts.

GOK Report

GOK is an open drain, active low signal that reports the fault events under the following conditions:

Over-current protection: when V_{CS} exceeds the CLREF threshold during normal operation, the GOK signal is pulled low after an 312 μ s response time.

Short-circuit protection: when load current reaches the I_{SC_LIMIT} rapidly, GOK is pulled low immediately.

Over-temperature protection: once the junction temperature exceeds 140°C, GOK is pulled low immediately.

Power MOSFET not fully on indication: after the power MOSFET is enabled and V_{OUT} begins ramping, after the max soft-start time (typical=200ms), if $V_{OUT} < 90\%$ of V_{IN} , GOK is pulled low immediately.

Power MOSFET gate has leakage indication:

at soft-start end, V_{GS} lower than V_{GS_TH} , the GOK is pulled low immediately or during normal operation, V_{GS} lower than V_{GS_TH} , the GOK is pulled low after an 200ms.

Above 5 faults occur, for latch off mode, GOK is pulled low and not release until the power recycling. For hiccup mode, after the retry delay time, the device turns on again through the soft-start process. When GOK is pulled low, the SS capacitor is discharged.

During soft-start, GOK is released once the UVLO is reached, therefore it can't be used directly as a Power Good indicator.

The HQUEF5016 can detect the internal power MOSFET drain-to-source, drain-to-gate short before the power MOSFET is turned on. If a short is detected, GOK is pulled low. One the short is removed, GOK is released high again.

The HQUEF5016 provides a comparator on the GOK pin to control the power MOSFET. If GOK drops below 0.3V, then the system sees this as a fault, and the power MOSFET turns off. FLT_TYPE outputs 0.3V to indicate GOK fault. The GOK fault is non-latch fault. Once GOK exceeds 1.4V, the device initiates a soft-start and resumes normal operation.

FLT_TYPE

To better distinguish the different fault, the HQUEF5016 provides FLT_TYPE with different voltage to indicate the different fault.

Table 4 indicates the indicated voltage of FLT_TYPE under different fault types.

The fault indication occurs after the power MOSFET is turned off, and indeed a fault has occurred. When multiple faults occur at the same time, the fault with high voltage will be displayed first, for example, OC and OTP occur at the same time, and FLT_TYPE will indicate 1.2V.

Table 4. FLT_TYPE under Different Fault Types

Fault Types	FLT_TYPE Voltage
OC	1.2V
SCP	1.5V
OTP/OVP	0.9V
GS/ DS Short	0.6V
charge pump NG or Gate Leakage	0.6V
GOK Fault	0.3V
Others	0.1V

This pin cannot be shorted to ground.

Parallel Application

The HQUEF5016 can be works in parallel operation to support higher current.

The HQUEF5016 current-sharing loop only functions during the start-up state. In the normal steady-state, the chip does not have active current-sharing function. Therefore, system design is necessary to ensure current sharing during steady-state conditions. Several points need to be noted: It's advisable to select chips from the same batch to ensure consistent $R_{DS(ON)}$ values. Pay attention to simulating the trace impedance for each path during the layout phase and make adjustments promptly. Also, consider the maximum system load current and the magnitude of overcurrent protection current. If the impedance of different phases in parallel applications is too large, it may cause the system to not function properly.

The following connections are required for multiple of HQUEF5016 in parallel operation. Please refer to the Typical Application Circuit section for more details.

- 1)Connect VIN to the same input bus.
- 2)Connect VOUT to the same output bus.
- 3)Tie ON together to realize power on sequence control.

4) Tie SS together to realize soft-start sequence control.

5) Tie GOK together for system fault control

6) Design the CS resistor (R_{CS}) with same resistance for each phase.

7) Tie IMON together and design the total system IMON resistor (R_{IMON}) as R_{CS}/N , where N is the phase count. This is to realize soft-start current

balance control.

8) Tie CLREF together.

9) Select the same MODE for all phases.

10) Tie VTEMP together to make the VTEMP always report the highest junction temperature of all phases.

APPLICATION INFORMATION

The HQUEF5016 is an electronic fuse designed for 4~16V system applications. It is used to manage inrush current and provide protection to downstream circuits and upstream bus systems in case of a fault condition.

Current Limit Reference Voltage (CLREF)

It is important to set an appropriate external overcurrent reference voltage for the circuit. When the voltage sampled from the CS(V_{CS}) exceeds the reference voltage(V_{CLREF}), it will trigger overcurrent protection. It is recommended to set the reference voltage within the range of 0.3-1.8V. A current of 10uA flows out from the CLREF pin to form a voltage across the resistor connected to ground, which provides the reference voltage. For example, when an R_{CS} resistor of 120k Ω is chosen, the reference voltage should be set to 1.2V.

Selecting the Circuit Breaker Resistor (R_{CS})

The overcurrent value (I_{OC_normal}) of HQUEF5016 should exceed the normal maximum load current (I_{LOAD_MAX}). Typically, the overcurrent is set within the range of 120% of I_{LOAD_MAX} or higher to provide a good margin of tolerance. I_{OC_normal} can be calculated using Equation.

$$I_{OC_normal} = \frac{V_{CLREF}}{R_{CS}} * 10^5 \text{ A}$$

For example, if the maximum system current is 50A and the reference voltage V_{CLREF} is 1.2V, then setting the overcurrent protection current I_{OC_normal} to 60A would correspond to a current-sense resistor R_{CS} with a resistance value of 2k Ω .

Current Monitor Setting (IMON)

R_{IMON} should be $\geq R_{CS}$. When the chip is used in multi-phase parallel configuration, the IMON pins of each phase circuit should be connected together. In cases where imbalances in phase currents occur during the start-up process due to differences in the chip itself or external factors, the chip's current sharing loop control comes into effect. It reduces the current in the phase that exceeds the average current, ensuring that each phase current is balanced and averaged.

UVLO Setting

If an application requires a higher VIN under-voltage lockout (UVLO) threshold, UVLO threshold can be programmed by external resistor divider. the rising UVLO threshold should be set sufficiently below the minimum input voltage, for this example, UV rising threshold is 9V. The UVLO threshold can be calculated by below equation

$$V_{UVLO_rising} = \frac{R_2 + R_1}{R_2} \times 1.4V$$

Assume R_1 is 100k Ω , R_2 can be computed by equation shown as

$$R_2 = \frac{R_1}{\frac{V_{UVLO_rising}}{1.4V} - 1} = 18.4k\Omega$$

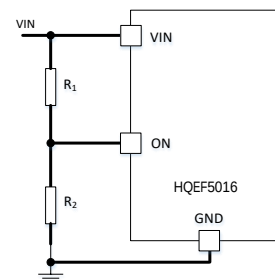


Figure 5: Adjustable UVLO

Power Supply Considerations

It is recommended to use a minimum capacitance of 0.1 μ F on the VIN pin of each device in parallel applications, close to the

device. This local capacitor can absorb spikes on the VIN pin during transient events such as hot-plug, short-circuit, and load discharge.

In addition, improving device performance and preventing negative output voltage deviation caused by parasitic inductance during the turn off process can be achieved by bypassing the device's output with a ceramic capacitor of 2.2 μ F or higher.

Under short-circuit and overload current conditions, when a device instantaneously interrupts a large amount of current, the input inductance generates positive voltage spikes at the input, while the output inductance generates negative voltage spikes at the output. The peak amplitude of these transient voltage spikes depends on the value of the inductance connected in series with the circuit. These transients can exceed the absolute maximum ratings of the device and may result in chip failures.

To protect the chip, it is recommended to add Transient Voltage Suppression (TVS) diodes at the input terminals to clamp the positive transient spikes at the input. Additionally, Schottky diodes can be used at the output terminals to absorb negative spikes.

PCB Layout Guideline

For minimum noise problem and best operating performance, the PCB is preferred to following the guidelines as reference.

1. For all applications, HQ recommends using a 0.1 μ F or larger ceramic decoupling capacitor between the VIN terminal and the GND terminal, placed near the chip pins.
2. For all applications, HQ recommends using a 2.2 μ F or larger ceramic decoupling capacitor between the VOUT terminal and the GND terminal, placed near the chip pins.
3. The power path for high current flow must be kept as short as possible, and the PCB copper traces must have dimensions capable of carrying at least twice the full load current.
4. The VIN and VOUT pins are used for heat dissipation. Connect them to as many copper areas as possible.
5. GND must be connected to the PCB ground plane at the IC terminal. The PCB ground must be a copper plane or copper island on the board.
6. Place a capacitor at least 1 μ F as close to VDD as possible.
7. Connect the other end of the components to the GND pin of the device and keep the trace length as short as possible. Minimize the trace length for C_{IN}, C_{OUT}, C_{VDD}, R_{CLREF}, R_{CS}, R_{IMON}, R_{SCREF}, R_{MODE}, and C_{SS} components to reduce parasitic effects on current limiting, overcurrent protection, and soft-start timing. These traces should not couple with any switching signals on the board.
8. Due to the direct control of device overcurrent protection behavior by IMON, CLREF, and CS pins, the PCB routing for these nodes should be kept away from any noisy (switching) signals.
9. Protection devices such as TVS, capacitors, or Schottky diodes must be physically close to the chip they are intended to protect. Route these protection devices with the shortest possible traces to minimize inductance.

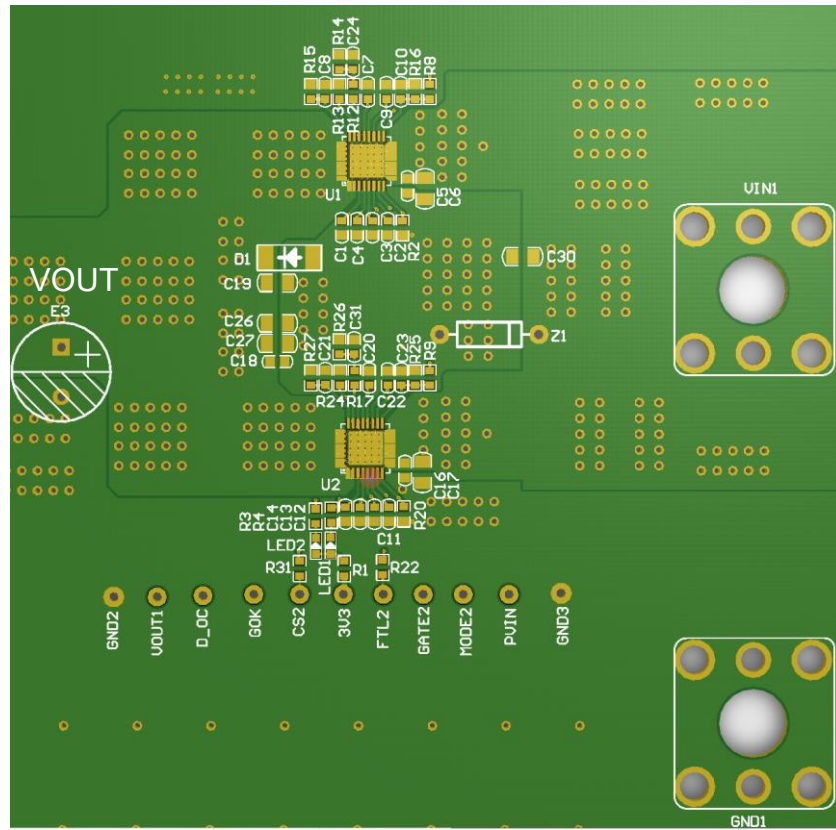


Figure 6: HQ EF5016 Two-Phase Parallel Application Layout Example

REFERENCE DESIGN

Note: Information in the following reference design sections is not part of HQ component specification. Customers are responsible for determining suitability of components chosen for their purposes and should validate their design implementation to make sure the proper system functionality.

Reference 1:

V_{IN} : 9V~14V

I_{LOAD} : 0~50A

Fault response: Hiccup Mode

C_{OUT} : 3000uF

T_{A_MAX} : 50°C

Output voltage slew rate: 1.2V/ms

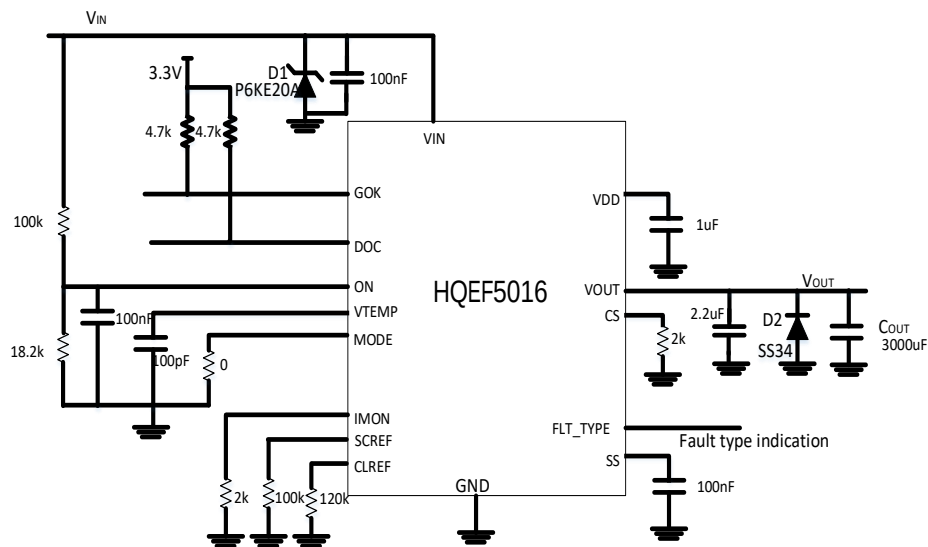


Figure 7: HQ EF5016 Single-Phase Application Circuit Diagram

Reference 2:

V_{IN} : 9V~14V

I_{LOAD} : 0~100A

Fault response: Hiccup Mode

C_{OUT} : 3000uF

T_{A_MAX} : 50°C

Output voltage slew rate: 1.2V/ms

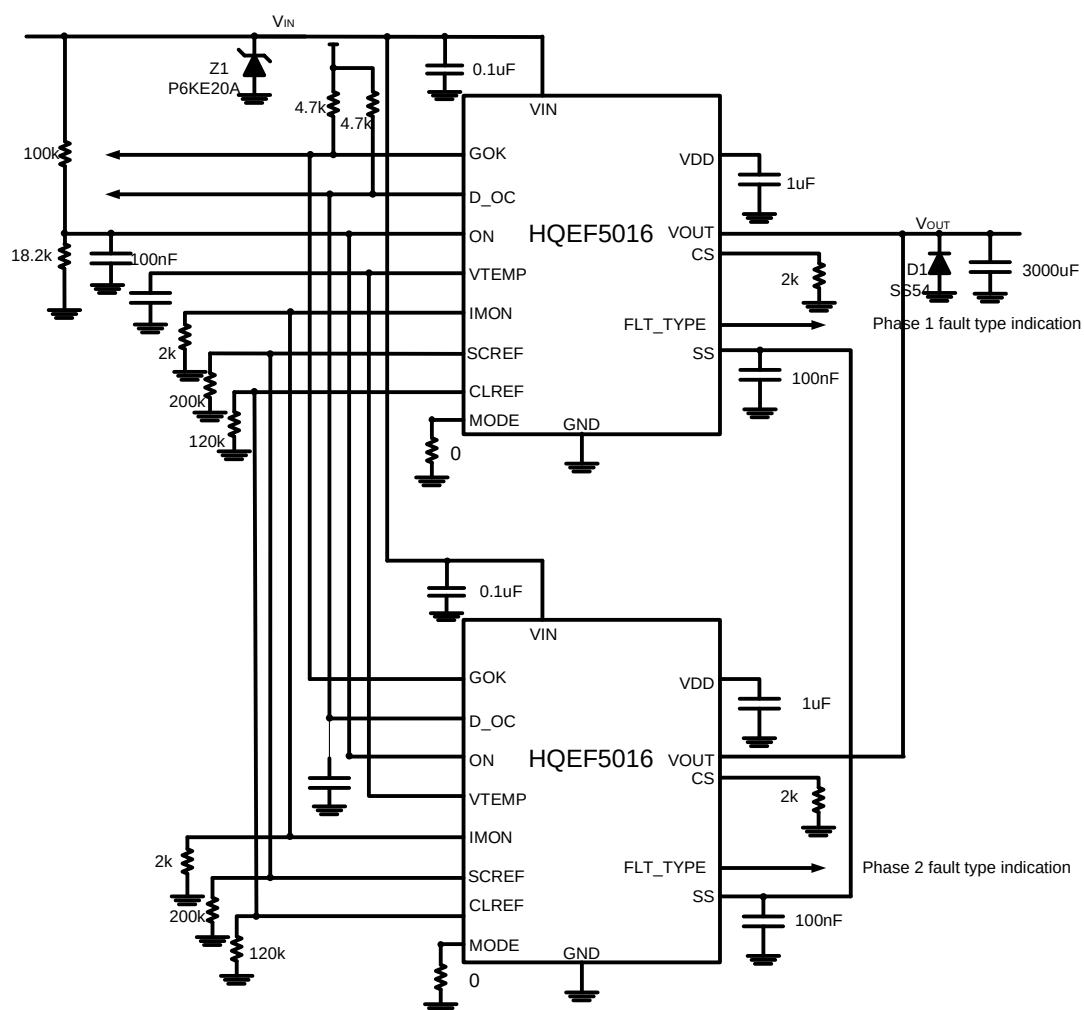
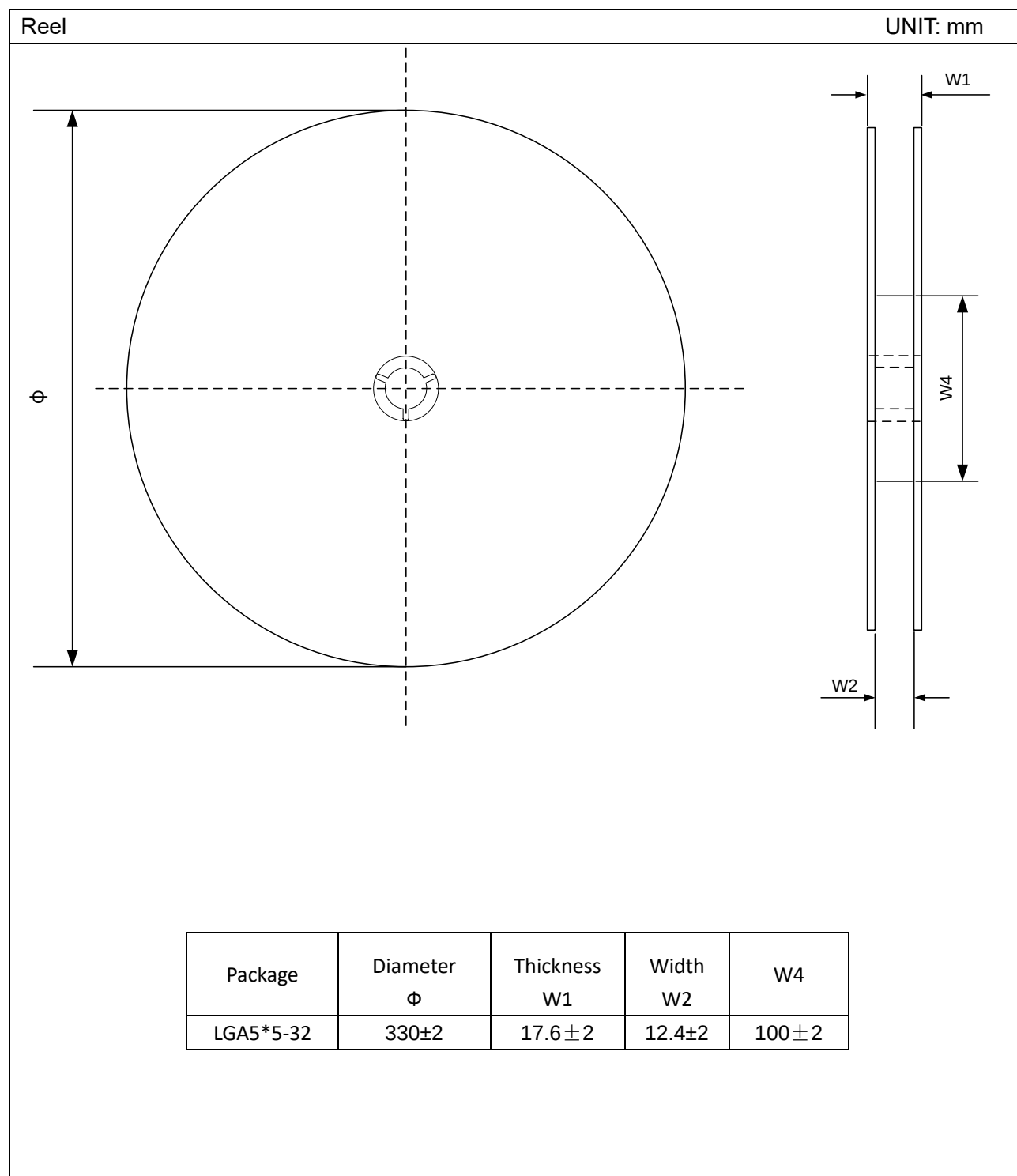
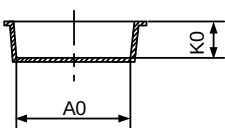


Figure 8: HQ EF5016 Two-Phase Application Circuit Diagram

TAPE AND REEL INFORMATION



UNIT: mm



SECTION B-B

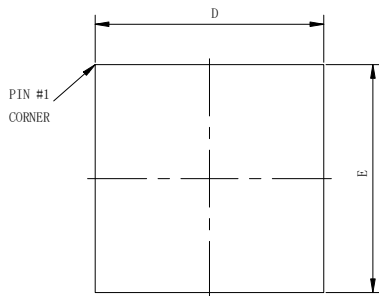
- 1) The carrier type is black, and colorless transparent.
- 2) Carrier camber is within 1mm in 100mm.
- 3) 10 pocket hole pitch cumulative tolerance: ± 0.20 .
- 4) All dimensions are in mm.

Package	Tape dimensions (mm)											
	P0	P2	P1	A0	B0	W	T	K0	Φ 1	Φ 2	E	F
LGA5*5-32	4.00±0.10	2.0±0.1	8.0±0.1	5.3±0.2	5.3±0.2	12±0.30	0.25±0.1	1.2±0.2	1.5±0.1	1.5±0.1	1.75±0.1	5.50±0.1

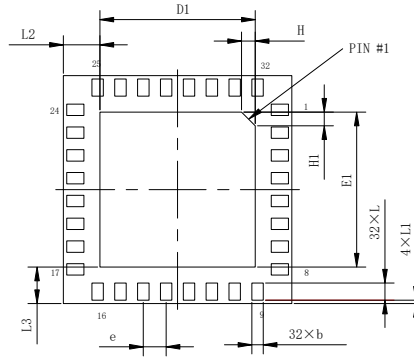
PACKAGE OUTLINE

LGA5*5-32

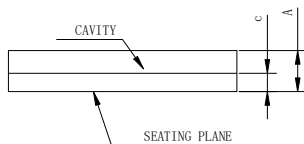
Unit: mm



Top View



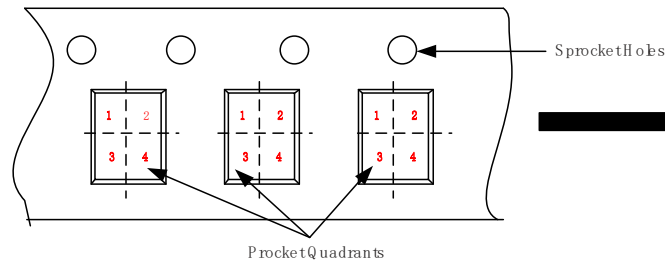
Bottom View



Side View

symbol	Dimension in mm		
	MIN	NOM	MAX
A	0.800	0.900	1.000
c	0.360	0.400	0.440
D	4.900	5.000	5.100
E	4.900	5.000	5.100
D1	3.300	3.400	3.500
E1	3.300	3.400	3.500
H	---	0.300	---
H1	---	0.300	---
L	0.325	0.375	0.425
L1	0.000	0.075	0.150
L2	0.725	0.800	0.875
L3	0.725	0.800	0.875
e	---	0.500	---
b	0.200	0.250	0.300

QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPAE



Package Type	Pin1 Quadrant
LGA5*5-32	1